

LUCAS CRUZ

Senior Data Engineer | Gen. AI Automation Engineer

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Curitiba, Brazil · Open to relocation across Europe · Requires visa sponsorship

PROFESSIONAL SUMMARY

Senior Data Engineer and Gen. AI Automation Engineer with 5+ years of experience building production data pipelines, cloud lakehouses, and LLM/n8n automation across IoT, fintech, agribusiness, and automotive intelligence. Proven track record delivering end-to-end data platforms, governed analytics datasets, conversational AI workflows, automated reporting systems, and customer-facing product implementation. Building a public portfolio of production AI platform systems (RAG, agents, evaluation, MCP, observability) over enterprise data through 2026. Fluent in English (C1); open to relocation across Europe.

TECHNICAL SKILLS

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Gen. AI & LLMs: LLM Integration, AI Agents (OpenAI, Claude, Gemini...), RAG, Prompt Engineering, Chatbot Development

Data Engineering, APIs & Orchestration: Python, SQL, NestJS, Next.js, FastAPI, SQLAlchemy, Pydantic, Apache Spark (PySpark), Pandas, dbt, Trino, REST APIs, Web Scraping, Parquet, Apache Airflow, n8n, Celery, Redis Streams, Meta Cloud API

Cloud & Lakehouses: Amazon Web Service (AWS), Microsoft Azure, Google Cloud (GCP), Oracle Cloud (OCI), Snowflake, Databricks, Amazon Redshift, Delta Lake, Oracle Autonomous Database

Databases & BI: PostgreSQL, SQL Server, MySQL, Redis, Power BI, Qlik Sense, Apache Superset, Looker

Security & Data Governance: LGPD/GDPR Compliance, Data Privacy, IAM, RBAC, Data Masking, Encryption (Rest/Transit), Audit Logging, RFID/NFC Tagging, Firewall

DevOps & Infrastructure: Docker, Kubernetes, Terraform, Azure Bicep, Azure Container Apps, Git, CI/CD Pipelines, Linux, Bash Scripting

Languages: Portuguese (Native) · English (C1 – Advanced) · Spanish (B1 – Intermediate)

WORK EXPERIENCE

Gen. AI Engineer — TK Technologies

Curitiba, PR – Brazil (Hybrid) · Feb 2026 – Present · 6-month contract, expected Aug 2026

- Architected a multi-channel CRM platform (WhatsApp, Instagram, Messenger) using NestJS, Next.js, and PostgreSQL logic, deploying to Azure Container Apps with Bicep IaC.
- Engineered a multi-source web scraping pipeline in Python targeting 10+ automotive parts websites, extracting pricing and product metadata at scale to deliver competitive market intelligence.
- Architected AI-powered n8n automation workflows (ETL, RPA, and chatbot agents) by integrating LLMs (Claude AI, Google AI Studio) and GCP APIs (Sheets, Drive) to enrich datasets and reduce manual analysis work.
- Developed a conversational AI chatbot that empowers business users to query and refine pricing datasets via natural language, reducing time-to-insight for non-technical stakeholders.
- Deployed containerised backend services on Azure using Docker, standardising infrastructure environments and improving deployment reliability.
- Coordinated Notion-based delivery sprints, customer-facing demos, product training, sales support, and post-launch support while preparing Play Store/EAS/DUNS mobile launch readiness.

Data Engineer — Spacecom Monitoramento

Curitiba, PR – Brazil (On-site) · Apr 2023 – Jul 2024

- Designed and maintained an enterprise Big Data Lakehouse on AWS S3 leveraging Medallion Architecture

(Bronze/Silver/Gold) alongside Delta Lake, Parquet, PySpark, and Trino to support heavy analytics workloads.

- Supported a high-availability IoT platform scaling to 95,000+ devices, efficiently processing hundreds of millions of telemetry records per day.
- Managed electronic security tag (RFID/NFC) data for government court systems, applying LGPD/GDPR-aligned privacy controls, access governance, and secure handling of sensitive judicial data.
- Orchestrated and optimised ETL pipelines with Apache Airflow to improve pipeline reliability; utilised Docker on AWS EC2 to reduce deployment issues.
- Remediated inconsistencies in pipeline steps to improve downstream analytics consistency, delivering reliable datasets to Qlik Sense and Apache Superset dashboards.

Data Engineer Consultant — wDiscover

Curitiba, PR – Brazil (Remote) · Jun 2022 – May 2024

- Delivered end-to-end data projects across the fintech, agribusiness, and operations sectors by integrating high-volume payment providers (PagSeguro, Stone, Cielo), John Deere IoT APIs, and REST APIs.
- Designed dimensional Data Warehouses on SQL Server and Oracle Autonomous Database; automated complex ETL workflows with Airflow on Docker/Oracle Cloud to reduce recurring manual effort.
- Built dynamic Power BI dashboards to track critical KPIs, financial metrics, and operational performance for diverse client domains.

Data Analyst — Itaete Group

Curitiba, PR – Brazil (Hybrid) · Jun 2020 – Feb 2023

- Designed a SQL Server Data Warehouse and robust Python/SQL pipelines using Pandas to cleanse, transform, and enrich data from IoT devices, PostgreSQL, APIs, and CSV sources.
- Built Power BI dashboards and automated legacy Excel/VBA reporting workflows, shortening data processing time and reducing manual operational workloads.
- Developed internal systems for inventory management and financial automation, streamlining invoice generation and automated weekly report distribution.

NOTABLE PROJECTS

WhatsApp CRM & Automation Platform

NestJS, Next.js, PostgreSQL, n8n, Azure Container Apps, Meta Cloud API, Docker · TK Technologies – Jun 2026 – Present

- Developed a production CRM receiving Meta Cloud API webhooks for WhatsApp, Instagram, and Messenger; integrated dynamic welcome/cooldown flows via PostgreSQL and Next.js agent inbox. Deployed on Azure ACA.

Automotive Market Price Intelligence Platform

Python, n8n, LLMs, GCP APIs, Azure, Docker · TK Technologies – Feb 2026 – Present

- Scraped 10+ automotive parts sites with Python; orchestrated AI agents via n8n with Claude AI, Google AI Studio, and GCP to automate price analysis. Fully containerised on Azure.

Telegram Conversational Assistant

n8n, LLMs, Telegram Bot API, Redis, Data Tables, FastAPI · TK Technologies – Feb 2026 – Present

- Built a private conversational automation workflow that answers from curated internal knowledge, logs user/conversation state, and triggers CDP demo jobs through integrated backend APIs.

IoT Data Lakehouse on AWS – Medallion Architecture

AWS S3, Apache Spark, Delta Lake, Airflow, Trino, Apache Superset · Client Project – Feb 2024

- Designed a Bronze/Silver/Gold lakehouse on AWS S3 ingesting from APIs and databases. Virtualised the Gold layer with Trino and surfaced governed datasets through BI dashboards.

AI Platform Engineering Portfolio (in progress — 2026)

FastAPI · PostgreSQL/pgvector · LangGraph · MCP · RAGAS · Docker · OpenTelemetry

- Building public, production-grade AI systems over enterprise data: RAG with hybrid retrieval and evaluation gates, LangGraph agent orchestration with human approval, MCP tool servers, and observability. Milestones shipping Jul–Dec 2026 on github.com/devzurc.

EDUCATION

Bachelor of Systems Analysis and Development — Dom Bosco University, Curitiba, Brazil · Jun 2023

Majors: Information Systems · Business Analytics · Data Analysis · Database Systems · Software Engineering

LICENSES & CERTIFICATIONS

- AI for Writing and Communicating — Google (Credential ID: HE8YE5IS7WWV) · Jul 2026
- AI for Research and Insights — Google (Credential ID: 4PZDF3XQOF55) · Jul 2026
- AI for Brainstorming and Planning — Google (Credential ID: 7SFW2HGOKIU5) · Jul 2026
- AI Fundamentals — Google (Credential ID: UKBGM0AZUIS2) · Jul 2026